From the discrete Bellman envelope to the sharp square-root causal entropy asymptotic

Research note for the finite-Shannon-theorem project

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Abstract

For a fixed finite family of finite positive probability laws with common entropy h, we derive the second-order coefficient of the minimum exact causal seed entropy directly from the discrete Bellman recursion. The argument does not assume a controlled central limit theorem or convergence of policies to a diffusion. A positive-time explicit profile has a summable one-step Taylor defect; order preservation propagates two translated barriers from the discontinuous initial condition. Uniform martingale tails then justify passage to the envelope mean. If the least varentropy is positive, the resulting entropy remainder is $O(n^{1/4}\log n)$. The zero-variance case is obtained by an explicit independent-noise comparison. These are mathematical proofs, not Lean certificates; the finite-tree Shannon sandwich is an input from the repository. No priority claim is made for the classical nonlinear heat profile.

1 Statement and precise scope

Let $\mathcal{A}$ be a nonempty finite action set. Each P_a is positive on a finite nonempty alphabet, and $H_2(P_a) = h$ for every a. At every round and history all actions remain available, with these same laws. A policy may use the entire observed history. Put

$$X_a = -\log_2 P_a(Y) - h, \quad Y \sim P_a, \qquad \sigma_a^2 = \mathbb{E}X_a^2, \quad b = \min_a \sigma_a, \quad a_+ = \max_a \sigma_a.$$

In the proofs write $a = a_+$. Let C_n be the infimum of the Shannon entropy, in bits, of an exact countable seed whose deterministic, history-consistent decoder works for every policy on the depth-n tree. Let L_n be the mean of the lower envelope of its policy surprisal CDFs. We use the finite theorem

$$L_n \le C_n \le L_n + c_*, \qquad c_* = \frac{1 + 1/\ln 2}{2}. \tag{1}$$

The same inequalities hold for the entropy of the finite greedy seed. For countable seeds the lower inequality follows by pushing the source to the finite tuple of policy transcripts and using entropy data processing; the upper inequality already has a finite witness.

Theorem 1. *Under these hypotheses,*

$$\boxed{C_n = nh + \sqrt{\frac{2}{\pi}}\,(a-b)\sqrt{n} + o(\sqrt{n}).} \tag{2}$$

The same statement holds for L_n and for the greedy seed entropy. If $b > 0$, all three quantities equal

$$nh + \sqrt{\frac{2}{\pi}}\,(a-b)\sqrt{n} + O(n^{1/4}\log n). \tag{3}$$

The implicit constant may depend on the fixed family. This is not a uniform estimate as $b \downarrow 0$, and it asserts no sharp logarithmic term when $a = b$.

The proof below applies more generally to any finite family of bounded, centered increment laws. Only the last use of (1) concerns surprisal. It does not address unequal action entropies, arbitrary history-dependent kernels, the QZR reserve constraint, or a horizon-dependent family of laws.

2 Exact discrete recursion

For a policy π, let $S_n^\pi = \sum_{k=1}^n X_{A_k}$, with the fresh increment drawn after the predictable choice A_k. Set

$$G_n(x) = \inf_\pi \mathbb{P}(S_n^\pi \le x), \qquad (Tf)(x) = \min_{j\in\mathcal{A}} \mathbb{E} f(x - X_j).$$

Condition on the first action and outcome. Continuation policies can be chosen separately for each outcome; finite horizon and finite alphabets ensure the minima are attained. Thus

$$G_0(x) = \mathbf{1}_{\{x\ge 0\}}, \qquad G_{n+1} = TG_n. \tag{4}$$

The operator is monotone, commutes with spatial translations and addition of constants, and satisfies $\|Tf - Tg\|_\infty \le \|f - g\|_\infty$. Moreover G_n is a CDF: it is the minimum of finitely many CDFs with bounded support. By the chain rule for transcript probabilities,

$$\frac{L_n - nh}{\sqrt{n}} = \int_0^\infty [1 - G_n(x\sqrt{n})]\, dx - \int_0^\infty G_n(-x\sqrt{n})\, dx. \tag{5}$$

There is no exchange of an infimum with an integral in this identity. The envelope is formed first, and then its mean is integrated.

3 A quantitative comparison for positive least variance

Assume $0 < b \le a$. Write $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$ and let Φ be its CDF. Define

$$F_{a,b}(z) = \begin{cases} \dfrac{2b}{a+b}\Phi(z/b), & z \le 0, \\ 1 - \dfrac{2a}{a+b}\Phi(-z/a), & z \ge 0. \end{cases} \qquad u(t,x) = F_{a,b}(x/\sqrt{t}), \quad t > 0. \tag{6}$$

Both values at zero are $b/(a+b)$. On either half-line, with $s = b$ on the negative side and $s = a$ on the positive side,

$$F'(z) = \frac{2}{a+b}\phi(z/s), \qquad F''(z) = -\frac{2z}{(a+b)s^2}\phi(z/s).$$

Consequently $F \in C^2(\mathbb{R})$, $F''(0) = 0$, and

$$u_t = \min_j \frac{\sigma_j^2}{2} u_{xx}. \tag{7}$$

Indeed, $u_{xx} \ge 0$ for $x < 0$, selecting b^2, and $u_{xx} \le 0$ for $x > 0$, selecting a^2; both sides vanish at zero. No classical third derivative at zero is required.

Fix $M = \max_j \|X_j\|_\infty$ and $\phi_0 = (2\pi)^{-1/2}$. Define

$$A = \frac{M^3\phi_0}{3(a+b)b^2}, \qquad B = \frac{a\phi_0}{2(a+b)}\big(3e^{-1/2} + 3\sqrt{3}\, e^{-3/2}\big),$$
$$\ell = \frac{2\phi_0}{a+b}, \qquad D = \frac{a\phi_0 e^{-1/2}}{a+b}.$$

Lemma 2 (One-step defect)**.** *For every $t > 0$,*

$$\|Tu(t,\cdot) - u(t+1,\cdot)\|_\infty \le At^{-3/2} + \frac{B}{2}t^{-2}. \tag{8}$$

Proof. On each open half-line,

$$F'''(z) = \frac{2}{(a+b)s^2}\big((z/s)^2 - 1\big)\phi(z/s).$$

Since $\sup_y |(y^2-1)\phi(y)| = \phi_0$ and F'' is continuous at zero, F'' is globally Lipschitz with constant $2\phi_0/((a+b)b^2)$. Taylor's integral remainder therefore gives, also across zero,

$$\left| \mathbb{E}u(t, x - X_j) - u(t,x) - \frac{\sigma_j^2}{2} u_{xx}(t,x) \right| \le At^{-3/2}.$$

The linear term vanishes because $\mathbb{E}X_j = 0$. Taking the minimum changes neither this bound nor its uniformity in x. Equation (7) identifies the quadratic part with u_t.

For $z = x/\sqrt{t}$,

$$u_{tt}(t,x) = \frac{3zF'(z) + z^2F''(z)}{4t^2}.$$

The elementary maxima $\sup_y |y|\phi(y) = \phi_0 e^{-1/2}$ and $\sup_y |y|^3\phi(y) = 3\sqrt{3}\phi_0 e^{-3/2}$ give $\|u_{tt}(t,\cdot)\|_\infty \le B/t^2$. Integrating this bound over a unit time step supplies the second remainder $B/(2t^2)$. □

Lemma 3 (Explicit CDF comparison)**.** *For $n \ge 1$, $m \ge 1$ and $R \ge 0$,*

$$\sup_z |G_n(z\sqrt{n}) - F_{a,b}(z)| \le 2\Phi(-R) + \frac{3A}{\sqrt{m}} + \frac{B}{m} + \ell a R\sqrt{\frac{m}{n+m}} + D\ln(1+m/n). \tag{9}$$

In particular the left side is $O(n^{-1/4}\sqrt{\log n})$ for $n \ge 2$.

Proof. Let $d = aR\sqrt{m}$ and $\delta = 2\Phi(-R)$. Directly from (6),

$$u(m, x-d) - \delta \le \mathbf{1}_{\{x\ge 0\}} \le u(m, x+d) + \delta.$$

Apply T^n, preserving order and the translations. By the contraction property and Lemma 2, the accumulated defect is at most

$$E_m := \sum_{k=0}^{n-1}\left(\frac{A}{(m+k)^{3/2}} + \frac{B}{2(m+k)^2}\right) \le \frac{3A}{\sqrt{m}} + \frac{B}{m}.$$

Thus $u(n+m, x-d) - \delta - E_m \le G_n(x) \le u(n+m, x+d) + \delta + E_m$. The spatial translation costs at most $\ell d/\sqrt{n+m}$. Also $\sup_x |u_t(t,x)| \le D/t$, so changing time from $n+m$ to n costs at most $D\ln(1+m/n)$. This proves (9). Choose $m = \sqrt{n}$, $R = \sqrt{\ln n}$, and use $2\Phi(-R) \le e^{-R^2/2}$ to obtain the stated order. □

4 The mean: uniform tails, an explicit error bound, and the coefficient

Under every policy, the increments are martingale differences in $[-M, M]$. Conditional Hoeffding's bound and iteration give $\mathbb{E}e^{\theta S_n^\pi} \le e^{n\theta^2M^2/2}$. Markov's inequality, optimized at $\theta = x/(M^2\sqrt{n})$ and applied also to $-S_n^\pi$, yields, for $x > 0$,

$$1 - G_n(x\sqrt{n}) \le e^{-x^2/(2M^2)}, \qquad G_n(-x\sqrt{n}) \le e^{-x^2/(2M^2)}. \tag{10}$$

These bounds are uniform in both n and the adaptive policy. The same bounds hold for $1 - F_{a,b}(x)$ and $F_{a,b}(-x)$ since $a \le M$. Dominated convergence in (5) is now justified. Moreover, if e_n denotes the right side of (9), then, for every $r > 0$,

$$\left|\frac{L_n - nh}{\sqrt{n}} - \mu_{a,b}\right| \le 2re_n + \frac{4M^2}{r}e^{-r^2/(2M^2)}, \tag{11}$$

where $\mu_{a,b}$ is the mean of $F_{a,b}$. Split the CDF-distance integral at $\pm r$ to obtain this inequality, using $\int_r^\infty e^{-x^2/(2M^2)}dx \le (M^2/r)e^{-r^2/(2M^2)}$. With $r = M\sqrt{2\ln n}$, Lemma 3 makes (11) $O(n^{-1/4}\ln n)$.

Finally, $\int_0^\infty \Phi(-x/s)\,dx = s/\sqrt{2\pi}$ gives

$$\mu_{a,b} = \frac{2a^2 - 2b^2}{(a+b)\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}}(a-b). \tag{12}$$

Equations (11), (12) and (1) prove both claims of Theorem 1 when $b > 0$. In particular, (9) and (11), plus $c_*/\sqrt{n}$, are an explicit finite-n bound for the normalized optimal entropy error.

5 Zero least variance: a controlled perturbation argument

Assume $b = 0 < a$. For $\varepsilon > 0$, replace every increment by $X_j + \varepsilon\xi$, where each fresh ξ is an independent symmetric sign. The controller is allowed to observe the pair of outcomes. The Bellman state is still the partial sum: the conditional future laws depend only on the next action, so observing the pair rather than its sum changes no value. Let G_n^ε be this envelope. Its extreme standard deviations are $a_\varepsilon = \sqrt{a^2 + \varepsilon^2}$ and $b_\varepsilon = \varepsilon$; the previous proof applies to this bounded, centered family.

For every $\eta > 0$ and $x \in \mathbb{R}$,

$$G_n^\varepsilon((x-\eta)\sqrt{n}) - \varepsilon^2/\eta^2 \le G_n(x\sqrt{n}) \le G_n^\varepsilon((x+\eta)\sqrt{n}) + \varepsilon^2/\eta^2. \tag{13}$$

To verify this, couple the original and perturbed sums using the same actions and original outcomes. Their difference is $\varepsilon\sum_{k=1}^n \xi_k$, whose tail beyond $\eta\sqrt{n}$ is at most ε^2/η^2. An original policy can be used in the perturbed system by ignoring the signs. Conversely, a perturbed policy can be run in the original system using independent auxiliary signs. Such randomization does not improve an infimum of event probabilities: conditional on its entire independent random tape it is a deterministic causal policy, so its event probability is an average of probabilities of admissible policies. These two comparisons, with shifted thresholds, prove (13).

First let $n \to \infty$ for fixed ε, η, then set $\eta = \sqrt{\varepsilon}$ and let $\varepsilon \downarrow 0$. The explicit profiles tend to the continuous CDF

$$F_{a,0}(x) = \begin{cases} 0, & x \le 0, \\ 2\Phi(x/a) - 1, & x > 0. \end{cases}$$

Equation (13) proves $G_n(x\sqrt{n}) \to F_{a,0}(x)$ at every x. The original-family bound (10) still holds, so dominated convergence gives mean $a\sqrt{2/\pi}$. This proves (2) for $b = 0 < a$, without asserting the remainder (3) in that case. If $a = b = 0$, every increment is zero almost surely, $G_n = G_0$, and (1) immediately gives (2). This finishes the proof.

6 Two consequences and a numerical specialization

Corollary 4 (Universality of asymptotically optimal seeds)**.** *Suppose $a > 0$. Let $J_n = -\log_2 w_n(S_n)$ be the self-information of an exact countable seed with finite entropy, and suppose $\mathbb{E}J_n - C_n = o(\sqrt{n})$. Then the law of $(J_n - nh)/\sqrt{n}$ converges in W_1 to the law with CDF $F_{a,b}$ (with the half-normal convention when $b = 0$).*

Proof. Every seed atom is no heavier than any transcript to which it is mapped. Thus its self-information stochastically dominates each policy transcript, and its CDF is at most their lower envelope. If Z_n has that envelope CDF, then $Z_n \preceq_{\mathrm{st}} J_n$, and $W_1(\mathcal{L}(J_n), \mathcal{L}(Z_n)) = \mathbb{E}J_n - L_n$. By (1) this is $o(\sqrt{n})$. The CDF convergence and uniform tail bound above give W_1 convergence of $(Z_n - nh)/\sqrt{n}$ to $F_{a,b}$. The triangle inequality proves the claim. □

The profile is not the limiting transcript law of one policy when $a > b$: every centered transcript has mean zero and the uniform tail bound forces any weak limit to have mean zero. The envelope profile has strictly positive mean. Different thresholds may need different minimizing policies. Treating the whole envelope as one policy law would invalidate the argument.

For $P = (1/2, 1/4, 1/4)$ and $Q = (q, q, 1-2q)$, let q be the unique root of $H_2(Q) = 3/2$ in $(0.410, 0.411)$. Entropy is strictly decreasing on this interval since its derivative is $2\log_2((1-2q)/q) < 0$. The endpoint signs are equivalent to the exact integer inequalities

$$100^{200} > 2^{300}41^{164}18^{36}, \qquad 1000^{2000} < 2^{3000}411^{1644}178^{356}.$$

Also $411^4 < 2^5 178^4$ implies $\sigma_Q^2 = 2q(1-2q)[\log_2(q/(1-2q))]^2 < 2(0.411)(0.18)(5/4)^2 < 1/4$. The companion script verifies these scalar integer comparisons exactly. Numerical evaluation gives

$$q \approx 0.41003242818198557608,$$
$$a = \sigma_P = 0.5, \qquad b = \sigma_Q \approx 0.45645077526396337302,$$
$$\sqrt{2/\pi}(a-b) \approx 0.03474725405181786408.$$

Hence this explicitly defined equal-entropy pair satisfies

$$C_n = \tfrac{3}{2}n + \sqrt{2/\pi}(\tfrac{1}{2} - \sigma_Q)\sqrt{n} + O(n^{1/4}\log n).$$

The displayed decimals are numerical evaluations, not interval certificates. The companion Python program checks integer-increment Bellman examples and the local-defect bound; it does not compute the optimal entropy or certify this nonlattice pair numerically.

Relation to prior work and verification status. Controlled central limit theorems and this nonlinear Gaussian profile are classical; see Peng [1] and Peng–Zhou [2], especially their one-sided tail formula. The self-contained contribution of this note to the project is the direct discrete comparison (9), the explicit mean-error estimate (11), and their connection to the already proved finite causal-seed sandwich. The argument has not been independently refereed or formalized in Lean. No claim is made that the probability result, the profile, or its coefficient is new in the literature.

References

[1] S. Peng. *A New Central Limit Theorem under Sublinear Expectations.* arXiv:0803.2656 (2008). https://arxiv.org/abs/0803.2656.

[2] S. Peng and Q. Zhou. *A hypothesis-testing perspective on the G-normal distribution theory.* arXiv:1909.03530 (2019), Theorem 2 and Corollary 1. https://arxiv.org/abs/1909.03530.

[3] *finite-Shannon-theorem*, source repository. https://github.com/jendamikulik/finite-Shannon-theorem. Finite inputs: `one_seed_shannon` and `entropy_ge_envelope`; the pre-existing `SecondOrder.lean` evaluates the profile mean and the transfer through a bounded entropy gap.